SAMYA SAMANTA

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EDUCATION

Ph.D. in Chemistry, Texas A&M University, College Station, TX
2021 – 2026
Thesis title: *Design and Application of Electrophilic Amine Salts for Modern Organic Synthesis*
Advisor: **David C. Powers** | GPA: 4.00 / 4.00

BS & MS in Chemistry, Indian Institute of Technology Kharagpur, India
2015 – 2020
GPA: 9.02 / 10.00

PROFESSIONAL SUMMARY

Ph.D. chemist specializing in synthetic organic chemistry, small molecule synthesis, method development, and reaction optimization. Designs robust synthetic protocols by integrating mechanistic insight with catalysis, electrochemistry, and photochemistry. Experienced in purification, analytical characterization, laboratory safety, technical documentation, and mentoring in research environments.

TECHNICAL SKILLS

- **Synthetic Chemistry**: Organic synthesis • Small molecule synthesis • Retrosynthetic analysis • Method development • Reaction optimization • Transition-metal catalysis • Electrochemistry • Photochemistry • Radical chemistry • Mechanistic investigations • Air-sensitive synthesis
- **Analytical & Characterization Techniques:** Column chromatography • HPLC • NMR • EPR • GC-MS • IR • UV-Vis • Cyclic voltammetry
- **Laboratory Operations:** Reaction setup • Purification • Compound characterization • Experimental troubleshooting • Protocol development • Technical documentation • Grant writing • Laboratory safety • Mentoring and training

RESEARCH EXPERIENCE

1. **Texas A&M University, College Station, TX, USA** *(PhD Candidate)*
   2021 – 2026
   - Developed three synthetic methodologies for aziridine synthesis, β-Phenethylamine synthesis, and a base-free C–N cross-electrophile coupling, resulting in publications in *JACS, Angew. Chem* and *Nat. Commun.*
   - Established a robust, scalable and reproducible synthetic protocol for *N*-pyridinium aziridines, a bench-stable synthon of aziridines, published in *Organic Syntheses.*
   - Developed two complementary aminyl radical generation platforms based on redox-active *N*-pyridinium and *N*-phenothiazinium amine salts, expanding the toolbox for *N*-centered radical generation.
   - Integrated electrochemistry, photochemistry, transition-metal catalysis, and mechanistic studies to develop and optimize new synthetic methodologies for C–N bond formation.
   - Mentored six undergraduate and graduate researchers in synthetic organic chemistry, analytical characterization, air-sensitive techniques, and laboratory safety, enabling independent research.
2. **Indian Institute of Technology Kharagpur, India** *(MS Researcher)*
   2019 – 2020
   - Performed extensive reaction optimization and substrate scope evaluation to develop organophotoredox-mediated amide and benzothiazole synthesis (*Chem. Eur. J.* **2020**, *26,* 3703–3708).
3. **Nanyang Technological University, Singapore** *(Research Intern)* ***May 2019 – July 2019***
   - Designed the synthesis and investigated the reactivity of osmium high-nuclearity carbonyl complexes for photoacoustic imaging (*J. Clust. Sci.* **2021**, *32*, 929–935).

4. **University of Victoria, BC, Canada** *(Research Intern)* ***May 2018 – July 2018***
   - Developed synthetic routes to carbazole-bis(tetrazole) ligands and evaluated the corresponding platinum and palladium complexes as sustainable oxidation catalysts (*J. Coord. Chem.* **2021,** *74,* 983–1008).
5. **Indian Institute of Technology Kanpur, India** *(Research Intern).* ***May 2017– July 2017***
   - Synthesized a hexatopic triptycene hexa-acid linker and investigated its metal-assisted self-assembly to generate honeycomb-structured metal–organic frameworks (MOFs).

SELECTED PUBLICATIONS

1. **Samanta, S.**; Patel, L.; Manna, D.; Ojeda, M.; Powers, D. C. Base-Free C–N Cross-Electrophile Coupling, *J. Am. Chem. Soc.* **2026**, *148,* 33104–33117.
2. **Samanta, S.**; Tan, H.; Chatterjee, S.; Powers, D. C. Synthesis of *N*-Pyridinium Aziridines. *Org. Synth.* **2026**, *103,* 78–100.
3. **Samanta, S.**; Ojeda, M.; Nguyen, P.; Powers, D. C. Reductive Activation of *N*-Phenothiazinium Amine Salts Affords *N*-Centered Radicals, *Tetrahedron Lett.* ***2026****, 183, 156204.*
4. **Samanta, S.**; Biswas, P.; O'Bannon, B.; Powers, D. C. *β*-Phenethylamine Synthesis: *N*-Pyridinium Aziridines as Latent Dual Electrophiles. *Angew. Chem. Int. Ed.,* **2024**, *63,* e202406335 (**Hot paper**). ***Top viewed article in 2025.***
5. Tan, H.; **Samanta, S.**; Maity, A.; Roychowdhury, P. Powers, D. C. *N*-Aminopyridinium Reagents as Traceless Activating Groups in the Synthesis of *N*-Aryl Aziridines. *Nat. Commun.,* **2022**, *13,* 3341. Highlighted in *Synform* **2022**/11, A169–A171 as literature coverage.

Complete publication list available on **Google Scholar**.

SELECTED POSTER PRESENTATIONS

1. **Samanta, S.**; Patel, L.; Manna, D.; Ojeda, M.; Powers, D. C. *Base-Free, Cross-Electrophile C–N Coupling with Exclusive Cross-Selectivity.*"
   a) **Gordon Research Conference on Organic Reactions and Processes** ***July 2025***
2. **Samanta, S.**; Powers, D. C. "*β-Phenethylamine Synthesis: N-Pyridinium Aziridines as Latent Dual Electrophiles.*"
   a) **Welch Conference** on 'Frontiers in Molecular Catalysis' ***Oct 2024***
   b) **Graduate Research Symposium** by ACS Division of Organic Chemistry at Virginia ***July 2024***
3. **Samanta, S.**; Powers, D. C. "*Development of 1,2-Aminofunctionalized Library from N-Pyridinium Aziridines.*"
   a) **The TexSyn Conference** at TAMU ***June 2024***
   b) **F.A. Cotton Medal for Excellence in Chemical Research Symposium** at TAMU ***April 2024***
   c) **FloHet-2024** at University of Florida ***March 2024***

ORAL PRESENTATIONS

1. **Samanta, S.**; Powers, D. C. "*N-Pyridinium Aziridines: Synthesis and Reactivity*." At **Pfizer Chemistry Connect Virtual Symposium** ***Nov 2024***
2. **Samanta, S.**; Powers, D. C. "*β-Phenethylamine Synthesis: N-Pyridinium Aziridines as Latent Dual Electrophiles.*" At **Southwest Regional Meeting of the American Chemical Society** ***Oct 2024***

HONORS & AWARDS

- Selected participant, **74th Lindau Nobel Laureate Meeting in Chemistry (2025)**

- **Derek and Christiane Barton Graduate Endowment (2025)**
- **Sharon Dabney Memorial Scholarship (2025)**
- **F. A. Cotton Endowed Memorial Graduate Travel Award (2025)**
- **Graduate Student Research and Presentation (RAP) Travel Award (2025)**
- **3rd place** in the poster presentation, **Welch Conference (2024)**
- **Grand Prize for the best poster** in **Joint Research Symposium, TAMU (2023**)
- **Martin Donald Corera Memorial Travel Award (2023)**

ORGANIZATIONAL LEADERSHIP AND VOLUNTEERING

1. **Phi Lambda Upsilon (Honorary Chemical Society), TAMU** *(Active member and ex-treasurer)* ***2022 – present***
   - Served as treasurer and chair of the judiciary board, managing financial operations and supporting the society's activities.
   - Volunteered for Chemistry Open House at Texas A&M University (2022 and 2023) to promote chemistry education and engage the broader Aggie community.
2. **ACS Division of Chemical Health and Safety Workshop** *(Moderator)*
   Aug 2023
   - Introduced the RAMP (recognize hazards, assess risk, minimize risk, and prepare for emergencies) paradigm through case studies and interactive discussions to develop effective laboratory safety practices.
3. **Graduate Student Association of Chemistry (GSAC), TAMU** *(At large representative)*
   2022 – 2023
   - Volunteered at various social and professional events to support, engage, and advocate for graduate students within the Chemistry department.
   - Mentored prospective graduate students during Graduate Recruiting Week (2022–2023), facilitating their transition and providing insights into the graduate experience.
4. **Technology Transfer Group, IIT Kharagpur** *(Coordinator)*
   2018 – 2019
   - Organized multiple events to promote intellectual property awareness and patent drive in IIT Kharagpur as the bridge between the research labs and industry.